

# KYLE J. UTLEY

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## Mechanical Engineering Student | Systems Design & Validation Engineering

### EDUCATION

#### The University of Alabama

Expected Graduation May 2028

Tuscaloosa, AL — Bachelor Of Science: Mechanical Engineering

- Cumulative GPA: 3.67 | Member, University of Alabama Honors College | Accelerated STEM-to-MBA Program

### RELEVANT EXPERIENCE

#### EQUIPMENT ENGINEERING CO-OP – LINE 1

Aug 2024 - Dec 2024 | Jan 2025 - Apr 2025

Honda Motors | Lincoln, AL

- Modeled a new structural lift-assist fixture in CATIA, applying DFM principles, then fabricated the frame in-house with a plasma cutter, lathe, and mill, joined via MIG welding; consolidated pneumatic controls, cut unit weight by 7.0 lb, and added rubber-cushioned motion limiters to reduce operator strain.
- Achieved zero unplanned downtime since installation and positive operator feedback, with a more serviceable design that simplified ongoing maintenance and troubleshooting.
- Diagnosed a recurring sensor jam as a mechanical design flaw, iterating actuator flag material (metal, nylon, Delrin) to eliminate wear; redesigned switches have had zero jams since installation, eliminating a manual-handling workaround that risked part damage and operator injury.
- Manufactured in-house replacement parts for worn equipment components, avoiding costly expedited outsourcing to external machine shops and meaningfully reducing part-replacement spend.

#### SUSPENSION TEAM MEMBER

Jan 2024 - Present

Crimson Racing – Formula SAE, The University of Alabama | Tuscaloosa, AL

- Designed a suspension linkage component (bellcrank) in SolidWorks for a formula-style vehicle platform, validating geometry via FEA against defined structural load cases and machining the final part from 7075-T6 aluminum to minimize mass while preserving structural margin.
- Component passed a full season of structural validation under real operating loads without a single failure, confirming design margin and manufacturing quality.
- Designed a precision-slotted rotational reference component and magnetic sensor mounting fixture to develop a new rotational speed detection method, manufacturing the part via waterjet to hold tight installation tolerances.
- Validated the new sensing method through extensive field testing, cross-referencing measured data against an existing reference system to confirm accuracy, establishing the method for use across multiple vehicle platforms.

### ADDITIONAL EXPERIENCE

#### STUDENT ASSISTANT

July 2026 - Present

Manderson Graduate School of Business, The University of Alabama | Tuscaloosa, AL

- Provide cross-functional support across reception, records management, event logistics, and communications, serving as a primary point of contact for visitors, students, and faculty.

### TECHNICAL SKILLS

#### CAD & Design

- SolidWorks | CATIA | FEA Analysis | Fusion 360
- Root Cause Analysis | DFMEA | DFM | Sensor Integration

#### Shop & Fabrication

- CNC Machining | 3D Printing | Waterjet | Plasma Cutter
- MIG Welding | Lathe | Mill | Press | Grinder | Bandsaw

### INVOLVEMENT AND HONORS

- Member, Alabama AI's Pals Mentorship Program
- Member, American Society of Mechanical Engineers
- University of Alabama Dean's List | Fall 2025
- University of Alabama President's List | Spring 2024